

1. [20 points]

Consider a clustering problem using K clusters and data $\{x_n \in \mathbb{R}^D\}_{n=1}^N$.

- (3 points) Formulate a unified optimization problem for clustering that (i) models fuzzy memberships and (ii) chooses each class representative to minimize the sum of L1/Manhattan distances between the representative and each data point in the associated class. The formulation should include a free parameter that allows us to tune the fuzziness of the clustering/memberships.

Solution:

Let $x_n \in \mathbb{R}^D$ for $n = 1, \dots, N$. Introduce fuzzy membership variables

$$u_{nk} \in [0, 1], \quad \sum_{k=1}^K u_{nk} = 1 \quad \text{for each } n.$$

Let $q > 1$ be a fuzziness parameter (larger q gives softer/fuzzier memberships). Choose class representatives $\mu_k \in \mathbb{R}^D$ and minimize the following objective:

$$J(u, \mu) = \sum_{n=1}^N \sum_{k=1}^K (u_{nk})^q d(x_n, \mu_k) = \sum_{n,k} (u_{nk})^q \sum_{d=1}^D |x_{nd} - \mu_{kd}|$$

subject to $u_{nk} \in [0, 1]$, $\sum_k u_{nk} = 1$ for all n . Here $d(\cdot, \cdot)$ denotes the Manhattan (L1) distance. The parameter $q > 1$ tunes fuzziness: as $q \downarrow 1$ memberships become crisper; as q increases, memberships become softer.

Rubric:

- 1 point: Correctly formulation of the objective using L1/Manhattan distances.
- 1 point: Includes a free parameter (e.g., $q > 1$) to tune fuzziness in the objective.
- 1 point: Correct constraints on fuzzy memberships ($\sum_k u_{nk} = 1$ and $u_{nk} \geq 0$).

- (10 points) Mathematically derive an efficient (without any brute-force/exhaustive searches, or stochastic/non-deterministic steps) implementable algorithm, with math-style pseudocode, to solve this optimization problem “very closely” (a slight modification in the objective function may be needed). Mathematically justify your algorithm.

Solution:

The original fuzzy L1 objective is hard to minimise directly because the L1 norm is nondifferentiable. We relax the coordinatewise absolute value by the Huber loss H_γ with parameter $\gamma > 0$:

$$H_\gamma(r) = \begin{cases} \frac{1}{2}r^2, & |r| \leq \gamma, \\ \gamma|r| - \frac{1}{2}\gamma^2, & |r| > \gamma. \end{cases}$$

Note $H_\gamma(r)$ is convex and differentiable everywhere; as $\gamma \downarrow 0$ it approaches $|r|$.

Define the relaxed objective

$$J_\gamma(u, \mu) = \sum_{n=1}^N \sum_{k=1}^K (u_{nk})^q \sum_{d=1}^D H_\gamma(\mu_{kd} - x_{nd}),$$

subject to $u_{nk} \in [0, 1]$ and $\sum_k u_{nk} = 1$ for each n . Here $q > 1$ is the fuzziness parameter.

We minimize J_γ by alternating minimization over u and μ , and use a schedule on γ (start large, reduce).

(Update memberships u) Fix μ . For each data point n define the (Huber) distance to cluster k

$$d_{nk}(\mu) := \sum_{d=1}^D H_\gamma(\mu_{kd} - x_{nd}) \geq 0.$$

For fixed μ the problem separates over n :

$$\min_{u_{n1}, \dots, u_{nK}} \sum_{k=1}^K (u_{nk})^q d_{nk} \quad \text{s.t.} \quad \sum_{k=1}^K u_{nk} = 1, \quad u_{nk} \geq 0.$$

Using a Lagrange multiplier gives

$$q(u_{nk})^{q-1} d_{nk} + \lambda_n = 0 \implies u_{nk} \propto d_{nk}^{-\frac{1}{q-1}}.$$

On normalizing, we get the update as,

$$u_{nk} = \frac{d_{nk}^{-\frac{1}{q-1}}}{\sum_{j=1}^K d_{nj}^{-\frac{1}{q-1}}}, \quad \text{if } d_{nj} > 0 \forall j.$$

If some $d_{nj} = 0$ (point coincides with a cluster center on all coordinates), set $u_{nj} = 1$ for such j and zero for others.

This update costs $O(KD)$ per data point to compute distances and memberships.

(Update cluster centers μ for fixed u) Fix u . Define weights $w_{nk} = (u_{nk})^q \geq 0$. The subproblem for cluster k is

$$\min_{\mu_k \in \mathbb{R}^D} J_k(\mu_k) \quad \text{where} \quad J_k(\mu_k) = \sum_{n=1}^N w_{nk} \sum_{d=1}^D H_\gamma(\mu_{kd} - x_{nd}).$$

This is separable in coordinates, so for each k and each coordinate d we solve the scalar convex problem

$$\min_{t \in \mathbb{R}} f(t) \quad \text{with} \quad f(t) = \sum_{n=1}^N w_{nk} H_\gamma(t - x_{nd}).$$

The derivative (using $\psi_\gamma(r) = H'_\gamma(r)$) is

$$f'(t) = \sum_{n=1}^N w_{nk} \psi_\gamma(t - x_{nd}), \quad \psi_\gamma(r) = \begin{cases} r, & |r| \leq \gamma, \\ \gamma \operatorname{sign}(r), & |r| > \gamma. \end{cases}$$

Because each term $\psi_\gamma(t - x_{nd})$ is monotone increasing in t , $f'(t)$ is monotone increasing and continuous; therefore f is convex and the unique minimizer is the root of $f'(t) = 0$ (or any point of the interval if flat).

Solver (Gradient Descent): Because the Huber loss is convex and its derivative is Lipschitz continuous, we can find the minimizer using gradient descent, which aligns well with the schedule for γ . Given a suitable step size (learning rate) $\alpha > 0$, we iteratively update each cluster center's coordinate:

$$\mu_{kd} \leftarrow \mu_{kd} - \alpha \sum_{n=1}^N w_{nk} \psi_\gamma(\mu_{kd} - x_{nd}).$$

We repeat this gradient descent step until the coordinates converge for the given γ .

Schedule on γ (to approach L1). To approximate the original L1 objective reliably, use a schedule:

$$\gamma_0 > \gamma_1 > \dots > \gamma_M > 0,$$

for example $\gamma_{t+1} = \eta \gamma_t$ with $\eta \in (0, 1)$ (e.g. $\eta = 0.5$ or 0.8). For each γ do alternating minimization (steps A,B) until the change is below tolerance; then reduce γ and continue, initializing μ and u from the previous stage. Starting with a large γ_0 makes the problem very smooth (quadratic-like); reducing γ progressively makes the solution closer to the L1 optimum.

Input: $\{x_n\}_{n=1}^N$, K , $q > 1$, $\{\gamma^{(s)}\}_{s=0}^M$, $\varepsilon > 0$, $T_{\max}$.

Initialize: $\mu_k^{(0)}$ (e.g. first K data points), $s \leftarrow 0$.

For $s = 0, \dots, M$ (schedule over $\gamma^{(s)}$):

Set $\gamma \leftarrow \gamma^{(s)}$.

Repeat for $t = 1, \dots, T_{\max}$:

(1) Compute $d_{nk} = \sum_{d=1}^D H_\gamma(\mu_{kd}^{(t-1)} - x_{nd})$.

(2) Update memberships $u_{nk}^{(t)} = \frac{d_{nk}^{-1/(q-1)}}{\sum_j d_{nj}^{-1/(q-1)}}$ (handle $d_{nj} = 0$ as above).

(3) $w_{nk}^{(t)} = (u_{nk}^{(t)})^q$.

(4) For each k, d find $\mu_{kd}^{(t)}$ by solving $\sum_n w_{nk}^{(t)} \psi_\gamma(\mu_{kd} - x_{nd}) = 0$

(5) If $\max_{k,d} |\mu_{kd}^{(t)} - \mu_{kd}^{(t-1)}| < \varepsilon$ then break.

End Repeat

Use final (μ, u) as initialization for next γ .

End For

Output: u, μ at the final $\gamma^{(M)}$.

Rubric (If using the same method):

- 2 points: Correct relaxed objective $J_\gamma(u, \mu)$ using Huber loss (or similar)
- 2 points: Correct membership update derivation (Lagrange multipliers)
- 2 points: Correct centers update method (Gradient descent)
- 2 points: Continuation schedule description (γ reduction)
- 2 points: Complete and accurate pseudocode integrating all steps

Rubric (Generalized for any valid approach):

- 2 points: **Handling L1 Non-differentiability:** For proposing a valid, deterministic strategy to handle the absolute value function (e.g., Huber loss relaxation or exact weighted-median formulation).
- 2 points: **Membership Update Derivation:** Correct derivation of the update rule for u_{nk} given fixed centers, enforcing the $\sum_k u_{nk} = 1$ constraint
- 2 points: **Center Update Derivation:** Correct derivation of the update rule for the cluster centers μ_k consistent with the chosen L1 strategy
- 2 points: **Mathematical Justification:** Sound theoretical justification for convergence (e.g., alternating minimization decreasing the objective) and an explanation of how it "very closely" solves the original fuzzy L1 problem.
- 2 points: **Pseudocode:** Complete and implementable pseudocode integrating all the steps. (only awarded if the previous updates were correct)

- (4 points) Suppose K was unknown, but only known to lie between $[3, 6]$, propose a strategy, based on “silhouette” analysis for choosing K . Give an implementable algorithm, with math-style pseudocode.

Solution:

For each datum x_i assigned to cluster A :

a_i = average distance between x_i and other points in cluster A

(Small a_i means cluster A is compact.)

$b_i = \min_{G \neq A} (\text{average distance between } x_i \text{ and points in cluster } G)$

(Large b_i means good separation from other clusters.)

The silhouette value is

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}, \quad -1 \leq s_i \leq 1.$$

For $K = 3, 4, 5, 6$:

1. Run the clustering algorithm with K clusters.
2. Obtain crisp labels $\hat{c}_i$.
3. For each data point x_i :
 Compute a_i .
 Compute b_i .
 Compute $s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$.
4. Compute average silhouette $\bar{s}_K = \frac{1}{N} \sum_{i=1}^N s_i$.

End For

Choose $K^* = \arg \max_{K \in \{3, 4, 5, 6\}} \bar{s}_K$.

If two values of K give similar $\bar{s}_K$, choose the smaller K (simpler model).

Rubric:

- 1 points: Correct definition of intra-cluster distance (a_i), inter-cluster distance (b_i) and silhouette formula (s_i)
- 2 points: Logic for averaging and selecting the maximum $\bar{s}_K$ and reasoning behind it (other strategies which were correct have also been awarded marks based on correctness)
- 1 point: Clear pseudocode for the overall algorithm

- (3 points) Describe and justify one practical benefit, and one practical disadvantage, for minimizing the sum of L1/Manhattan distances, instead of the usual formulation.

Answer the same questions, i.e., benefit and disadvantage, if we replace L1 distances by L2 distances.

Solution:

For minimizing L1 norm:

- **Practical benefit:** Robustness to outliers. L1 distance penalizes large deviations linearly (not quadratically), so a few extreme points do not dominate the objective; weighted medians (component-wise) are less influenced by outliers than means.
- **Practical disadvantage:** Centers update requires solving a weighted-median problem (no simple vector-average closed form). This results in slightly more costly updates (sorting or selection required) compared to the simple weighted average for L2. Also the objective is non-differentiable at points where coordinates coincide, which rules out using some gradient-based solvers

For replacing L1 norm with L2 norm

- **Practical benefit:** Closed-form centers update: for weighted squared L2 loss the minimizer is the weighted mean $\mu_k = (\sum_n w_{nk} x_n) / (\sum_n w_{nk})$. This gives very cheap $O(ND)$ updates (no sorting) and a smooth objective function.
- **Practical disadvantage:** Sensitivity to outliers. Squaring large deviations gives outliers disproportionate influence on centers and memberships.

Rubric:

- 1.5 points: For L1 norm benefit and disadvantage (0.75 each)
- 1.5 points: For L2 norm benefit and disadvantage (0.75 each)

2. [20 points]

Consider a clustering problem using K clusters and data $\{x_n \in \mathbb{R}^D\}_{n=1}^N$.

• (20 points) Describe an implementable algorithm, with math-style pseudocode, for solving the K-means clustering problem using simulated annealing. Describe the initialization, the update steps, and the termination criterion.

Solution:

1. Energy function (5 points)

Data: $x_n \in \mathbb{R}^D$, $n = 1, \dots, N$. Binary labels $u_{nk} \in \{0, 1\}$ with $\sum_{k=1}^K u_{nk} = 1$ for each n .

Given labels $\{u_{nk}\}$ the centroid of cluster k is

$$\mu_k = \frac{\sum_{n=1}^N u_{nk} x_n}{\sum_{n=1}^N u_{nk}} \quad (\text{if } n_k := \sum_n u_{nk} > 0).$$

The K-means energy (within-cluster sum of squared distances) is

$$E(u) = \min_{\{\mu_k\}} \sum_{k=1}^K \sum_{n=1}^N u_{nk} \|x_n - \mu_k\|^2 = \sum_{k=1}^K \sum_{n: u_{nk}=1} \|x_n - \mu_k\|^2.$$

2. Temperature (2 points)

Introduce a temperature scalar $T > 0$. Intuition: T controls the probability of accepting an energy *increase* in order to escape local minima. Two limiting behaviours:

- Large T : $\exp(-\Delta E/T)$ approaches 1 $\Rightarrow$ high probability of accepting increases, leading to more exploration.
- $T \rightarrow 0$: $\exp(-\Delta E/T)$ approaches 0 $\Rightarrow$ accept only non-increasing moves (greedy K-means).

3. Proposal of a candidate labeling (5 points)

Use single-point reassignment updates:

- Randomly pick an index n .
- Let a be the current cluster of x_n (so $u_{n,a} = 1$).
- Pick a target cluster $b \neq a$ uniformly at random (or choose the second-best nearest centroid).
- Candidate labeling: move x_n from a to b (i.e. set $u_{n,a} = 0, u_{n,b} = 1$).

Efficiently compute energy change, Maintain for each cluster:

$$n_k = \sum_n u_{nk}, \quad S_k = \sum_n u_{nk} x_n, \quad Q_k = \sum_n u_{nk} \|x_n\|^2.$$

Then for cluster k ,

$$E_k = \sum_{n: u_{nk}=1} \|x_n - \mu_k\|^2 = Q_k - \frac{\|S_k\|^2}{n_k}.$$

If we move x_n from a to b , the only changed clusters are a, b . Their post-move statistics are

$$\begin{aligned} n'_a &= n_a - 1, & S'_a &= S_a - x_n, & Q'_a &= Q_a - \|x_n\|^2, \\ n'_b &= n_b + 1, & S'_b &= S_b + x_n, & Q'_b &= Q_b + \|x_n\|^2. \end{aligned}$$

Compute

$$\begin{aligned} E_{\text{old}}^{(ab)} &= \left(Q_a - \frac{\|S_a\|^2}{n_a} \right) + \left(Q_b - \frac{\|S_b\|^2}{n_b} \right), \\ E_{\text{new}}^{(ab)} &= \left(Q'_a - \frac{\|S'_a\|^2}{n'_a} \right) + \left(Q'_b - \frac{\|S'_b\|^2}{n'_b} \right). \end{aligned}$$

Then

$$\boxed{\Delta E = E_{\text{new}}^{(ab)} - E_{\text{old}}^{(ab)}}.$$

4. Accept-reject criterion (6 points)

Given ΔE and temperature $T > 0$, decide whether to accept the candidate move using the following:

$$P(\text{accept}) = \begin{cases} 1 & \text{if } \Delta E \leq 0 \\ \exp\left(-\frac{\Delta E}{T}\right) & \text{if } \Delta E > 0 \end{cases}$$

Draw a random number $r \sim \text{Uniform}(0, 1)$. If $r < P(\text{accept})$, apply the move and update $u_{n\cdot}, n_k, S_k, Q_k$ to the primed values. If rejected, keep the current labeling and statistics.

Justification:

- When $\Delta E \leq 0$, the move is always accepted (greedy descent), improving or keeping the objective.
- When $\Delta E > 0$, allowing probabilistic increases helps escape shallow local minima. A sequence of accepted uphill moves may later lead to a larger decrease.
- As T is lowered, the probability of accepting uphill moves drops; when T reaches 0, we recover pure K-means local descent.

5. Temperature schedule (2 points)

Using an exponential schedule:

$$T_{t+1} = \alpha T_t, \quad 0 < \alpha < 1 \quad (\text{e.g. } \alpha = 0.9 \text{ or } 0.95).$$

Choices and termination:

- Set initial T_0 to be a rough scale of typical increases (e.g. mean squared distance to initial centroids).
- Terminate when $T < T_{\min}$ (small threshold), or after a fixed number of temperature steps, or if no moves are accepted for several temperature steps.

Algorithm (compact math-style pseudocode)

Initialize: u_{nk} (random or K-means++), compute $n_k, S_k, Q_k, T \leftarrow T_0$.

Repeat until termination:

for $m = 1, \dots, M$ (inner loop):

pick n uniformly, $a \leftarrow \text{cluster}(n)$, pick $b \neq a$.

compute ΔE as above.

if $(\Delta E \leq 0)$ or $(\text{Uniform}(0, 1) < \exp(-\Delta E/T))$ then

accept and update $u_{n\cdot}, n_k, S_k, Q_k$.

$T \leftarrow \alpha T$.

Output: final labels and centroids $\mu_k = S_k/n_k$.

Rubric:

- 5 points: describe energy function for binary labels u_{nk}
- 2 points: introduce temperature
- 5 points: describe a way to propose a candidate labeling
- 6 points: describe accept-reject criteria
- 2 points: describe temperature update schedule

3. [20 points]

- (4 points) Mathematically formulate a sparse-dictionary learning problem on 9×9 patches in 512×512 X-ray CT image (of attenuation coefficients) as a unified optimization problem using L_q (quasi) norm based penalty on the dictionary coefficients for $q < 1$, and enforcing non-negativity constraints (hard) on the dictionary atoms and the coefficients.

Solution:

Using an L_q quasi-norm penalty with $0 < q < 1$ we formulate

$$\min_{D, \{c_i\}} \sum_{i=1}^M \left\{ \|x_i - Dc_i\|_2^2 + \alpha \|c_i\|_q^q \right\} \quad \text{s.t.} \quad D \geq 0, \quad c_i \geq 0 \quad \forall i, \quad \|d_k\|_2 \leq 1 \quad \forall k,$$

where $\|c\|_q^q = \sum_{j=1}^K |c_j|^q$ (quasi-norm), $\alpha > 0$ controls sparsity, and the constraints enforce nonnegativity and norm bounds on dictionary atoms (to avoid scale ambiguity).

Rubric:

- 2 points: correct unified objective with L_q penalty and nonnegativity constraints.
- 1 point: norm constraint on dictionary atoms to avoid scale ambiguity.
- 1 point: clear specification of patches and variables.

- (6 points) Describe an implementable algorithm for solving the sparse dictionary-learning problem. Describe the update steps for all variables clearly mathematically up to implementation-level detail.

Solution:

We solve the optimization problem from part (a):

$$\min_{D, \{c_i\}} \sum_{i=1}^M (\|x_i - Dc_i\|_2^2 + \alpha \|c_i\|_q^q)$$

subject to

$$D \geq 0, \quad c_i \geq 0, \quad \|d_k\|_2 \leq 1 \quad \text{for all atoms } d_k.$$

Since the problem is not jointly convex in D and $\{c_i\}$, we use **alternating projected gradient descent**.

Initialization:

- Initialize dictionary $D^{(0)} \geq 0$ with random nonnegative entries.
- Normalize each column $d_k^{(0)}$ so that $\|d_k^{(0)}\|_2 \leq 1$.
- Initialize $c_i^{(0)} \geq 0$ (e.g., zeros).

Step 1: Update coefficients c_i (fix D).

For each patch x_i , minimize

$$f_i(c) = \|x_i - Dc\|_2^2 + \alpha \sum_{j=1}^K c_j^q.$$

For $c_j > 0$, the gradient is

$$\nabla_c f_i(c) = 2D^\top (Dc - x_i) + \alpha q \begin{bmatrix} c_1^{q-1} \\ \vdots \\ c_K^{q-1} \end{bmatrix}.$$

Projected gradient step:

$$\tilde{c}_i = c_i^{(t)} - \eta_c \nabla_c f_i(c_i^{(t)}),$$

$$c_i^{(t+1)} = \max(0, \tilde{c}_i) \quad (\text{elementwise}).$$

Repeat this inner update for a few iterations for each i .

Step 2: Update dictionary D (fix $\{c_i\}$).

We minimize

$$\sum_{i=1}^M \|x_i - Dc_i\|_2^2.$$

Stacking $X = [x_1, \dots, x_M]$ and $C = [c_1, \dots, c_M]$, this becomes

$$\|X - DC\|_F^2.$$

Gradient with respect to D :

$$\nabla_D = 2(DCC^\top - XC^\top).$$

Gradient step:

$$\tilde{D} = D^{(t)} - 2 * \eta_D (D^{(t)}CC^\top - XC^\top).$$

Projection:

(a) Enforce nonnegativity:

$$\tilde{D} \leftarrow \max(0, \tilde{D})$$

(elementwise).

(b) For each column k ,

$$d_k \leftarrow \begin{cases} \frac{\tilde{d}_k}{\|\tilde{d}_k\|_2}, & \text{if } \|\tilde{d}_k\|_2 > 1, \\ \tilde{d}_k, & \text{otherwise.} \end{cases}$$

Full Algorithm (Pseudocode):

Input: $X, K, q, \alpha, \eta_c, \eta_D, T_{\max}$
Initialize $D^{(0)}, C^{(0)} \geq 0$
for $t = 0, \dots, T_{\max} - 1$:
 (A) Update C :
 for each i :
 $g = 2D^\top (Dc_i - x_i) + \alpha q c_i^{q-1}$
 $c_i \leftarrow \max(0, c_i - \eta_c g)$
 (B) Update D :
 $G = 2(DCC^\top - XC^\top)$
 $D \leftarrow \max(0, D - \eta_D G)$
 Normalize columns if needed
end for

Rubric:

- 4 points: correct projected gradient descent update for dictionary atoms d_k , including correct gradient expression and explicit projection (nonnegativity + ℓ_2 normalization).
- 2 points: correct projected gradient descent update for coefficients c_i , including correct gradient expression and explicit nonnegativity projection.

- (4 points) Formulate a unified optimization problem for performing Bayesian image reconstruction for a X-ray CT image x given acquired Radon-transform data y , the CT-imaging matrix A , and a dictionary based prior model using a given dictionary D , a L_q -norm based sparsity prior on the dictionary coefficients, and enforcing non-negativity constraints (hard) on the image x and the dictionary coefficients.

Solution:

Assume zero-mean i.i.d. Gaussian noise in the Radon domain.

The unified optimization problem is:

$$\min_{x, \{c_i\}} \sum_{i=1}^M \|R_i x - D c_i\|_2^2 + \alpha \sum_{i=1}^M \|c_i\|_q^q + \beta \|Ax - y\|_2^2$$

subject to

$$x \succeq 0, \quad c_i \succeq 0 \text{ for all } i.$$

Here:

- A is the CT system matrix,
- R_i extracts the i -th 9×9 patch from x ,
- D is the given dictionary,
- $\|c_i\|_q^q = \sum_j |c_{ij}|^q$, with $0 < q \leq 1$,
- $\alpha, \beta > 0$ are regularization parameters.

Rubric:

- 2 points: correct unified objective (data term + patch term + L_q penalty).
- 1 point: nonnegativity constraints on x and c_i .
- 1 point: correct use of A , R_i , and dictionary D .

- (6 points) Describe an implementable algorithm for solving the image re- construction problem. Describe the update steps for all variables clearly mathematically up to implementation-level detail..

Solution:

We solve

$$\min_{x, \{c_i\}} \sum_{i=1}^M \|R_i x - D c_i\|_2^2 + \alpha \sum_{i=1}^M \|c_i\|_q^q + \beta \|Ax - y\|_2^2$$

subject to

$$x \succeq 0, \quad c_i \succeq 0.$$

We use alternating projected gradient descent.

Step 1: Update image x (fix $\{c_i\}$).

Define

$$F(x) = \sum_{i=1}^M \|R_i x - D c_i\|_2^2 + \beta \|Ax - y\|_2^2.$$

Gradient with respect to x :

$$\nabla_x F = 2 \sum_{i=1}^M R_i^\top (R_i x - D c_i) + 2\beta A^\top (Ax - y).$$

Gradient step:

$$\tilde{x} = x - \eta_x \nabla_x F.$$

Projection (nonnegativity):

$$x \leftarrow \max(0, \tilde{x}) \quad (\text{elementwise}).$$

Step 2: Update coefficients c_i (fix x).

For each patch i , minimize

$$f_i(c_i) = \|R_i x - D c_i\|_2^2 + \alpha \|c_i\|_q^q.$$

Gradient:

$$\nabla_{c_i} = 2D^\top (D c_i - R_i x) + \alpha q c_i^{q-1}.$$

Gradient step:

$$\tilde{c}_i = c_i - \eta_c \nabla_{c_i}.$$

Projection:

$$c_i \leftarrow \max(0, \tilde{c}_i) \quad (\text{elementwise}).$$

Full Algorithm:

Input: $y, A, D, R, q, \alpha, \beta, \eta_x, \eta_c, T_{\max}$
Initialize $x^{(0)} \geq 0, c_i^{(0)} \geq 0$ for all i .
for $t = 0, \dots, T_{\max} - 1$:
 (A) Update x :
 $\nabla_x = 2 \sum_{i=1}^M R_i^\top (R_i x - D c_i) + 2\beta A^\top (Ax - y)$
 $x \leftarrow \max(0, x - \eta_x \nabla_x)$
 (B) Update c_i for each i :
 $\nabla_{c_i} = 2D^\top (D c_i - R_i x) + \alpha q c_i^{q-1}$
 $c_i \leftarrow \max(0, c_i - \eta_c \nabla_{c_i})$
end for

Rubric:

- 4 points: correct gradient for x and correct projection $x \geq 0$.
- 2 points: correct gradient for c_i and correct projection $c_i \geq 0$.

4. [20 points]

Let y be a noisy image with N pixels having values $y_i \in \mathbb{R}$ at pixel i .

Let $y_i := x_i + \eta_i$ where x_i is the (unknown) noiseless value and $\eta_i \sim \mathcal{N}(0, \sigma^2 I)$ is the noise.

Let $\mathcal{N}$ denote the 4-neighborhood system. For each $(i, j) \in \mathcal{N}$, we introduce a binary variable $z_{ij} \in \{0, 1\}$ where $z_{ij} = 1$ indicates an edge between pixels i and j .

Let $\gamma > 0$ be a fixed constant (free parameter).

The prior on random field Z is

$$p(Z) \propto \exp \left(-\gamma \sum_{(i,j) \in \mathcal{N}} z_{ij} \right).$$

Let $w_0 > w_1 \geq 0$ be fixed constants (free parameters).

The conditional prior distribution on random field X given Z is

$$p(X | Z) \propto \exp \left(- \sum_{(i,j) \in \mathcal{N}} \frac{w_{ij}}{2} (X_i - X_j)^2 \right),$$

where

$$w_{ij} = \begin{cases} w_0, & \text{if } z_{ij} = 0, \\ w_1, & \text{if } z_{ij} = 1. \end{cases}$$

• (3 points) Mathematically derive and write an implementable mathematical expression (up to an additive constant) for the negative log-posterior

$$E(x, z; y) := -\log p(x, z | y).$$

Solution:

Using Bayes' rule:

$$p(x, z | y) \propto p(y | x) p(x | z) p(z).$$

Likelihood:

$$p(y | x) \propto \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - x_i)^2 \right).$$

Prior on Z :

$$p(z) \propto \exp \left(-\gamma \sum_{(i,j) \in \mathcal{N}} z_{ij} \right).$$

Conditional prior on X :

$$p(x | z) \propto \exp \left(- \sum_{(i,j) \in \mathcal{N}} \frac{w_{ij}}{2} (x_i - x_j)^2 \right).$$

Therefore, up to an additive constant,

$$E(x, z; y) = \frac{1}{2\sigma^2} \sum_i (y_i - x_i)^2 + \sum_{(i,j) \in \mathcal{N}} \frac{w_{ij}}{2} (x_i - x_j)^2 + \gamma \sum_{(i,j) \in \mathcal{N}} z_{ij}.$$

Rubric:

- 1 point: correct likelihood
- 1 point: correct priors
- 1 point: correct combined expression

- (7 points: 3 + 4) For a fixed z and y , is $E(x; z, y)$ a convex function of x ? Prove or disprove. Also, derive a closed-form mathematical expression (implementable) of x that minimizes $E(x; z, y)$.

Solution:

For fixed z (so w_{ij} are fixed) and fixed y , the x -dependent part of the negative log-posterior is

$$E(x; z, y) = \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - x_i)^2 + \frac{1}{2} \sum_{(i,j) \in \mathcal{N}} w_{ij} (x_i - x_j)^2 + \text{constant (w.r.t. } x).$$

Compute first and second derivatives entrywise.

The gradient components are, for each pixel i ,

$$\frac{\partial E}{\partial x_i} = \frac{1}{\sigma^2} (x_i - y_i) + \sum_{j: (i,j) \in \mathcal{N}} w_{ij} (x_i - x_j).$$

The Hessian $H \in \mathbb{R}^{N \times N}$ therefore has entries

$$H_{ii} = \frac{1}{\sigma^2} + \sum_{j: (i,j) \in \mathcal{N}} w_{ij}, \quad H_{ij} = \begin{cases} -w_{ij}, & (i, j) \in \mathcal{N}, \\ 0, & \text{otherwise.} \end{cases}$$

Let $v \in \mathbb{R}^N$ be any vector. Then the quadratic form is

$$v^\top H v = \frac{1}{\sigma^2} \sum_{i=1}^N v_i^2 + \sum_{(i,j) \in \mathcal{N}} w_{ij} (v_i - v_j)^2.$$

(The second term equals $\sum_i \sum_{j: (i,j) \in \mathcal{N}} \frac{1}{2} w_{ij} (v_i - v_j)^2$ and is nonnegative.)

Because $w_{ij} \geq 0$ and $\sigma^2 > 0$, we have for any nonzero v

$$v^\top H v \geq \frac{1}{\sigma^2} \sum_i v_i^2 > 0.$$

Hence H is positive definite, so $E(x; z, y)$ is *strictly convex* in x .

$E(x; z, y) \text{ is strictly convex in } x \text{ (for fixed } z, y).$

Rubric:

- 1 point: correct gradient and Hessian entries
- 2 points: show $v^\top H v > 0$.

Because E is strictly convex, the unique minimizer x^* is obtained by solving

$$\nabla_x E(x^*; z, y) = 0.$$

Writing the gradient equation componentwise gives, for each i ,

$$\frac{1}{\sigma^2} (x_i^* - y_i) + \sum_{j: (i,j) \in \mathcal{N}} w_{ij} (x_i^* - x_j^*) = 0.$$

Rearrange to obtain a linear system $Ax^* = b$ with

$$A_{ii} = \frac{1}{\sigma^2} + \sum_{j: (i,j) \in \mathcal{N}} w_{ij}, \quad A_{ij} = -w_{ij} \quad \text{for } (i, j) \in \mathcal{N},$$

$$b_i = \frac{1}{\sigma^2} y_i.$$

Thus the minimizer is

$x^* = A^{-1}b$

Rubric:

- 1 point: set gradient to 0
- 2 points: derive A_{ii}, A_{ij}, b_i correctly
- 1 point: state $x^* = A^{-1}b$ or any other method

- (4 points) For a fixed x and y , derive a closed-form mathematical expression (implementable) of z that minimizes $E(z; x, y)$.

Solution:

For fixed x , the z -dependent part of the objective is

$$E(z; x, y) = \sum_{(i,j) \in \mathcal{N}} \left[\frac{w_{ij}}{2} (x_i - x_j)^2 + \gamma z_{ij} \right].$$

Minimize independently for each (i, j) .

If $z_{ij} = 0$:

$$E_0 = \frac{w_0}{2} (x_i - x_j)^2.$$

If $z_{ij} = 1$:

$$E_1 = \frac{w_1}{2} (x_i - x_j)^2 + \gamma.$$

Choose $z_{ij} = 1$ iff $E_1 \leq E_0$:

$$\frac{w_1}{2} (x_i - x_j)^2 + \gamma \leq \frac{w_0}{2} (x_i - x_j)^2.$$

This gives threshold:

$$(x_i - x_j)^2 \geq \frac{2\gamma}{w_0 - w_1}.$$

$$z_{ij}^* = \begin{cases} 1 & \text{if } (x_i - x_j)^2 \geq \frac{2\gamma}{w_0 - w_1}, \\ 0 & \text{otherwise.} \end{cases}$$

Rubric:

- 2 points: correct analysis using 2 cases for z_{ij}
- 2 points: correct final threshold rule

- (6 points: 3 + 3) Consider the alternating minimization algorithm:

(a) Given $z^{(t)}$, compute $x^{(t+1)} := \arg \min_x E(x; z^{(t)}, y)$.

(b) Given $x^{(t+1)}$, update each $z^{(t+1)} := \arg \min_z E(z; x^{(t+1)}, y)$.

Prove that the alternating optimization algorithm converges.

Is the alternating optimization algorithm guaranteed to converge to the global minimum ?

Solution:

Each step performs exact minimization:

$$\begin{aligned} E(x^{(t+1)}, z^{(t)}) &\leq E(x^{(t)}, z^{(t)}), \\ E(x^{(t+1)}, z^{(t+1)}) &\leq E(x^{(t+1)}, z^{(t)}). \end{aligned}$$

Thus,

$$E^{(t+1)} \leq E^{(t)}.$$

Since $E(x, z; y)$ is bounded below,

$E^{(t)}$ converges.

Rubric:

- 3 points: monotone decrease argument

The energy is:

- convex in x for fixed z ,
- but non-convex jointly in (x, z) (because z is discrete).

Thus alternating minimization converges to a stationary point, but is NOT guaranteed to reach the global minimum.

Rubric:

- 3 points: correct statement about local vs global minimum with explanation